

Candidate Name _____

Centre Number

Candidate
Number

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CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

PHYSICS

PAPER 4 A2 Core

9702/4

OCTOBER/NOVEMBER SESSION 2002

1 hour

Candidates answer on the question paper.
No additional materials.

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You may lose marks if you do not show your working or if you do not use appropriate units.

FOR EXAMINER'S USE

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This question paper consists of 15 printed pages and 1 blank page.



Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2} QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = Ar\eta v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

Answer **all** the questions in the spaces provided.

- 1 A kettle is rated as 2.3 kW. A mass of 750 g of water at 20 °C is poured into the kettle. When the kettle is switched on, it takes 2.0 minutes for the water to start boiling. In a further 7.0 minutes, one half of the mass of water is boiled away.

(a) Estimate, for this water,

(i) the specific heat capacity,

specific heat capacity = J kg⁻¹ K⁻¹

(ii) the specific latent heat of vaporisation.

specific latent heat = J kg⁻¹
[5]

- (b) State **one** assumption made in your calculations, and explain whether this will lead to an overestimation or an underestimation of the value for the specific latent heat.

.....
.....
.....[2]

- 2 Fig. 2.1 gives information on three lines observed in the emission spectrum of hydrogen atoms.

wavelength/nm	photon energy / 10^{-19} J
656	3.03
486	
1880	1.06

Fig. 2.1

- (a) Complete Fig. 2.1 by calculating the photon energy for the wavelength of 486 nm.

[2]

- (b) Fig. 2.2 is a partially completed diagram to show energy levels of a hydrogen atom.

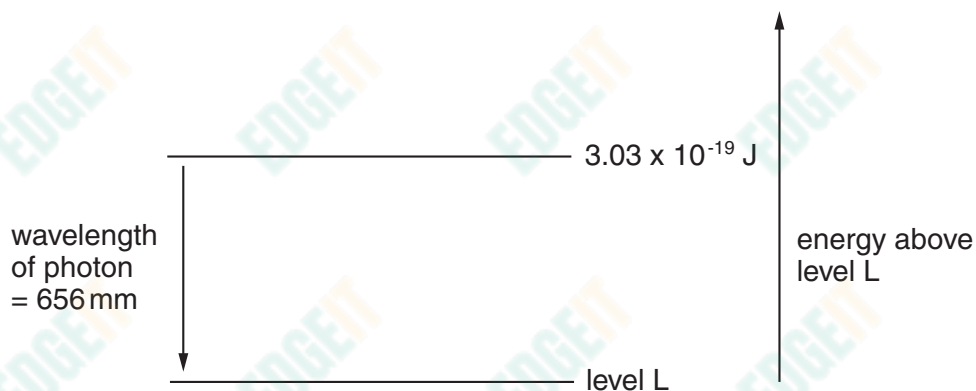


Fig. 2.2

On Fig. 2.2 draw one further labelled energy level, and complete the diagram with arrows to show the energy changes for the other two wavelengths. [3]

- 3 A student sets out to investigate the oscillation of a mass suspended from the free end of a spring, as illustrated in Fig. 3.1.

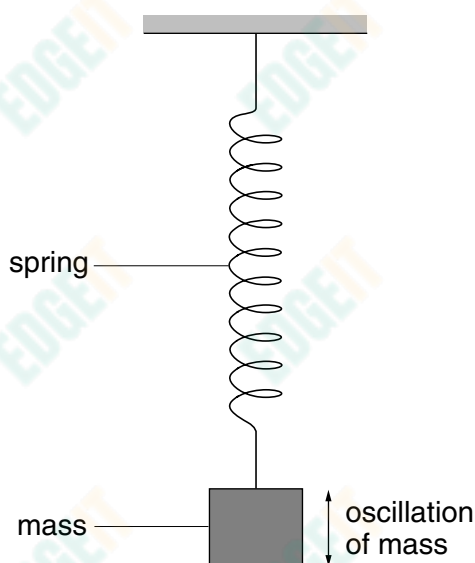


Fig. 3.1

The mass is pulled downwards and then released. The variation with time t of the displacement y of the mass is shown in Fig. 3.2.

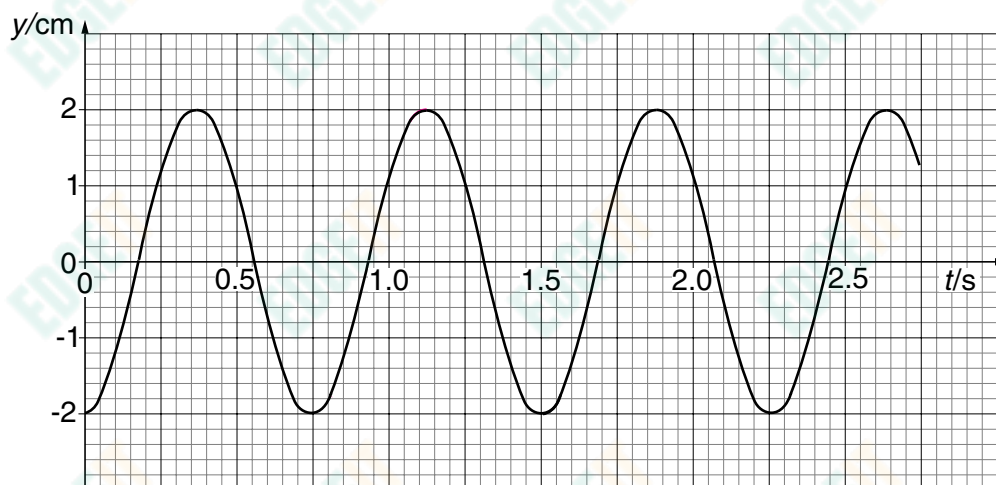


Fig. 3.2

- (a) Use information from Fig. 3.2
- (i) to explain why the graph suggests that the oscillations are undamped,

.....

- (ii) to calculate the angular frequency of the oscillations,

angular frequency = rad s^{-1}

- (iii) to determine the maximum speed of the oscillating mass.

speed = m s^{-1}
[6]

- (b) (i) Determine the resonant frequency f_0 of the mass-spring system.

f_0 = Hz

- (ii) The student finds that if short impulsive forces of frequency $\frac{1}{2}f_0$ are impressed on the mass-spring system, a large amplitude of oscillation is obtained. Explain this observation.

.....
.....
.....
[3]

- 4 If an object is projected vertically upwards from the surface of a planet at a fast enough speed, it can escape the planet's gravitational field. This means that the object can arrive at infinity where it has zero kinetic energy. The speed that is just enough for this to happen is known as the escape speed.

- (a) (i) By equating the kinetic energy of the object at the planet's surface to its total gain of potential energy in going to infinity, show that the escape speed v is given by

$$v^2 = \frac{2GM}{R},$$

where R is the radius of the planet and M is its mass.

- (ii) Hence show that

$$v^2 = 2Rg,$$

where g is the acceleration of free fall at the planet's surface.

[3]

- (b) The mean kinetic energy E_k of an atom of an ideal gas is given by

$$E_k = \frac{3}{2} kT,$$

where k is the Boltzmann constant and T is the thermodynamic temperature.

Using the equation in (a)(ii), estimate the temperature at the Earth's surface such that helium atoms of mass 6.6×10^{-27} kg could escape to infinity.

You may assume that helium gas behaves as an ideal gas and that the radius of Earth is 6.4×10^6 m.

temperature = K [4]

- 5 Some capacitors are marked '48 μ F, safe working voltage 25 V'.

Show how a number of these capacitors may be connected to provide a capacitor of capacitance

- (a) 48 μ F, safe working voltage 50 V,

[2]

- (b) 72 μ F, safe working voltage 25 V.

[2]

- 6 (a) A charged particle may experience a force in an electric field and in a magnetic field.

State two differences between the forces experienced in the two types of field.

1.
.....

2.
.....[4]

- (b) A proton, travelling in a vacuum at a speed of $4.5 \times 10^6 \text{ m s}^{-1}$, enters a region of uniform magnetic field of flux density 0.12 T. The path of the proton in the field is a circular arc, as illustrated in Fig. 6.1.

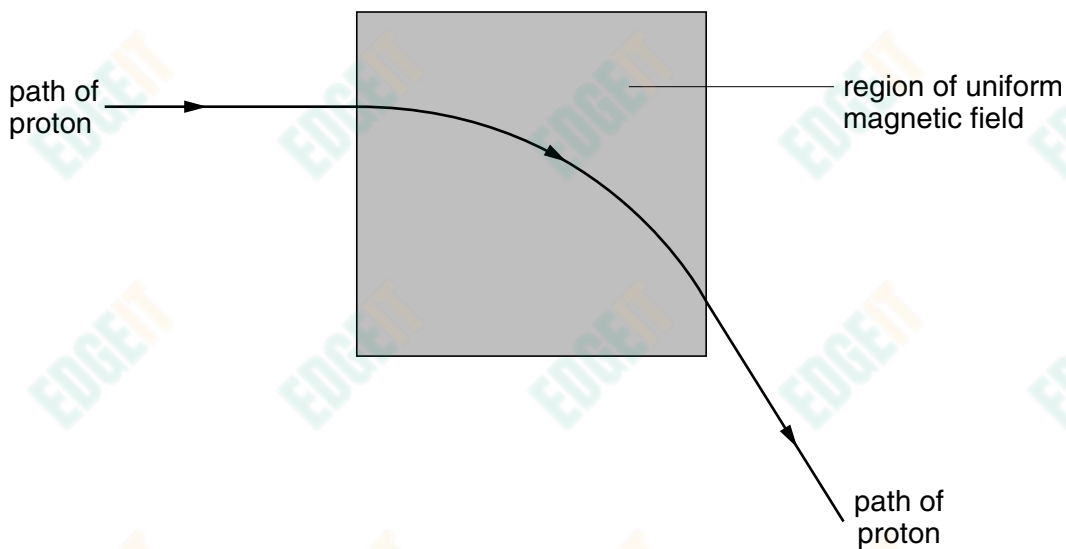


Fig. 6.1

- (i) State the direction of the magnetic field.

.....

- (ii) Calculate the radius of the path of the proton in the magnetic field.

radius = m
[4]

- (c) A uniform electric field is now created in the same region as the magnetic field in Fig. 6.1, so that the proton passes undeviated through the region of the two fields.
- (i) On Fig. 6.1 mark, with an arrow labelled E , the direction of the electric field.
- (ii) Calculate the magnitude of the electric field strength.

field strength = V m^{-1}
[3]

- (d) Suggest why gravitational forces on the proton have not been considered in the calculations in (b) and (c).

.....
.....[1]

- 7 A metal wire is held taut between the poles of a permanent magnet, as illustrated in Fig. 7.1.

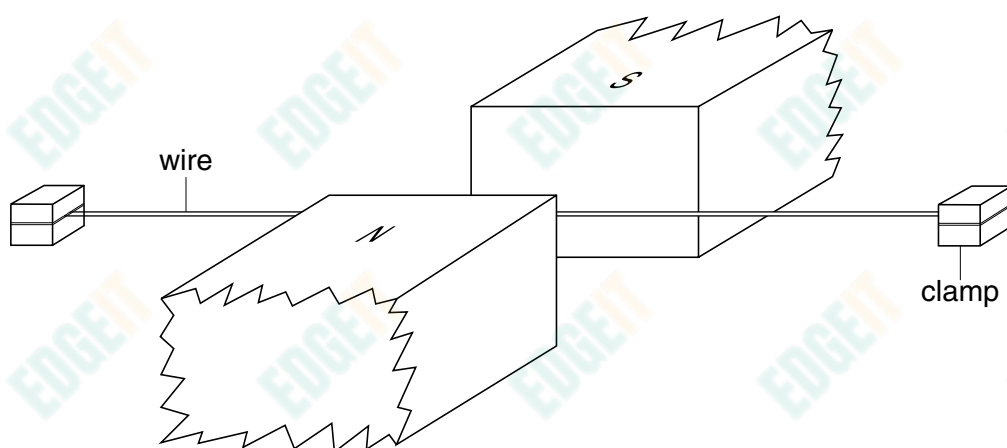


Fig. 7.1

A cathode-ray oscilloscope (c.r.o.) is connected between the ends of the wire. The Y-plate sensitivity is adjusted to 1.0 mV cm^{-1} and the time base is 0.5 ms cm^{-1} .

The wire is plucked at its centre. Fig. 7.2 shows the trace seen on the c.r.o.

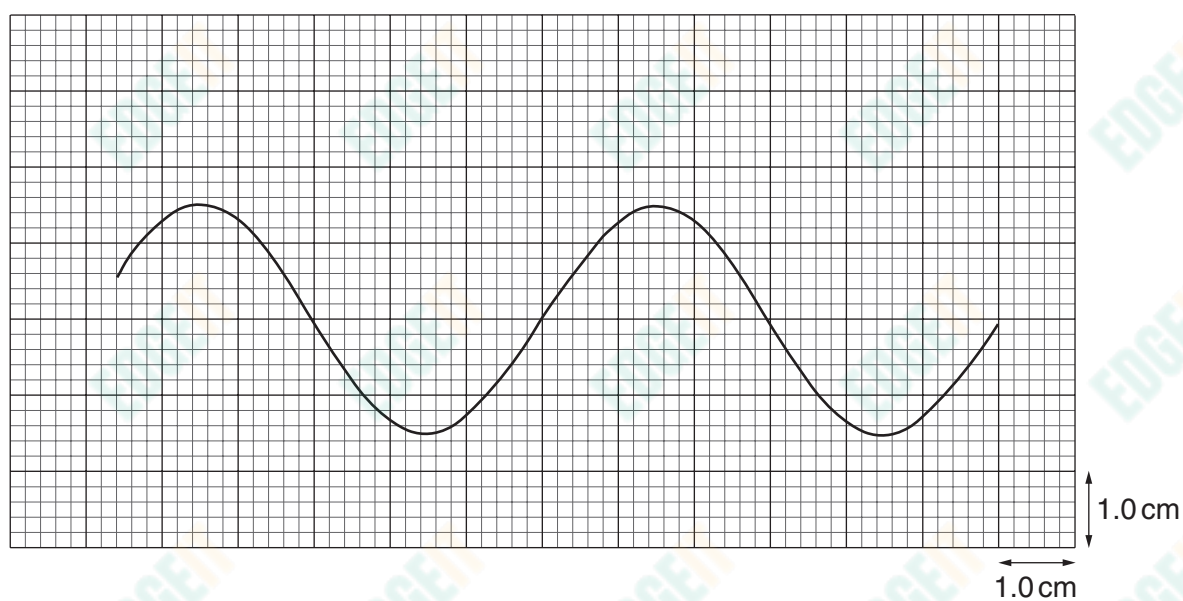


Fig. 7.2

(a) Making reference to the laws of electromagnetic induction, suggest why

(i) an e.m.f. is induced in the wire,

.....

.....

.....

(ii) the e.m.f. is alternating.

.....

.....

.....

[4]

(b) Use Fig. 7.2 and the c.r.o. settings to determine the equation representing the induced alternating e.m.f.

equation: [4]

- 8 (a) Define the term radioactive *decay constant*.

.....
.....
.....[2]

- (b) State the relation between the activity A of a sample of a radioactive isotope containing N atoms and the decay constant λ of the isotope.

.....[1]

- (c) Radon is a radioactive gas with half-life 56 s. For health reasons, the maximum permissible level of radon in air in a building is set at 1 radon atom for every 1.5×10^{21} molecules of air. 1 mol of air in the building is contained in 0.024 m^3 .

Calculate, for this building,

- (i) the number of molecules of air in 1.0 m^3 ,

number =

- (ii) the maximum permissible number of radon atoms in 1.0 m^3 of air,

number =

- (iii) the maximum permissible activity of radon per cubic metre of air.

activity = Bq
[5]

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